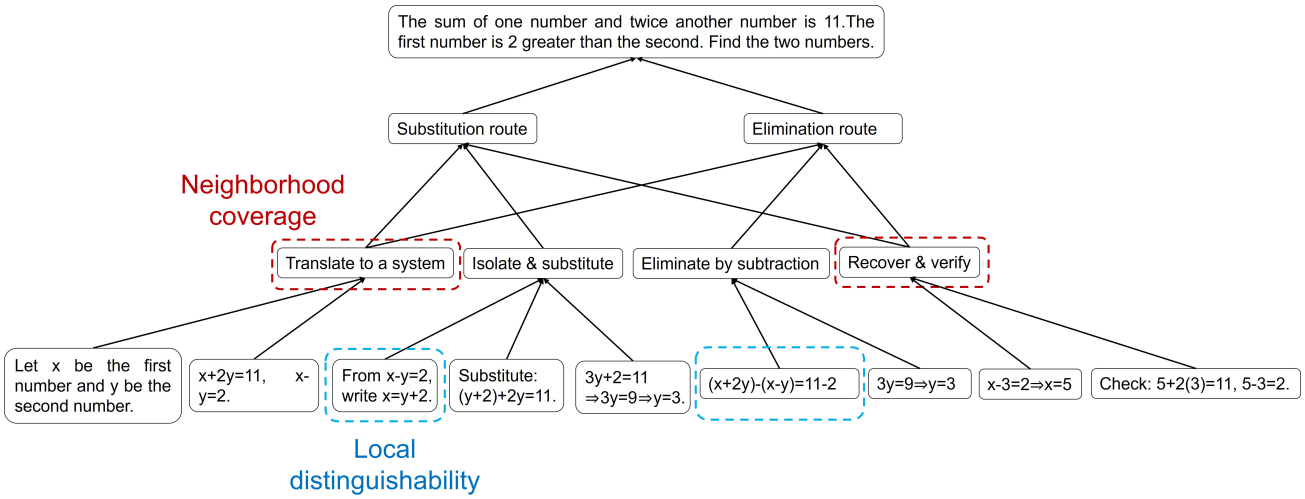

Figure H6. **A concrete algebra example illustrating skills, routing mechanisms, and identification conditions.** Shared modules such as *"Translate to a system"* and *"Recover & verify"* demonstrate neighborhood coverage. By contrast, branch-specific local traces such as *"write* $x = y + 2$*"* and *"*$(x + 2y) - (x - y)$*"* make the higher-level route choice locally distinguishable. Restricted choice sets are reflected in the fact that, at each intermediate state, only a small set of next modules or trace continuations is algebraically plausible.

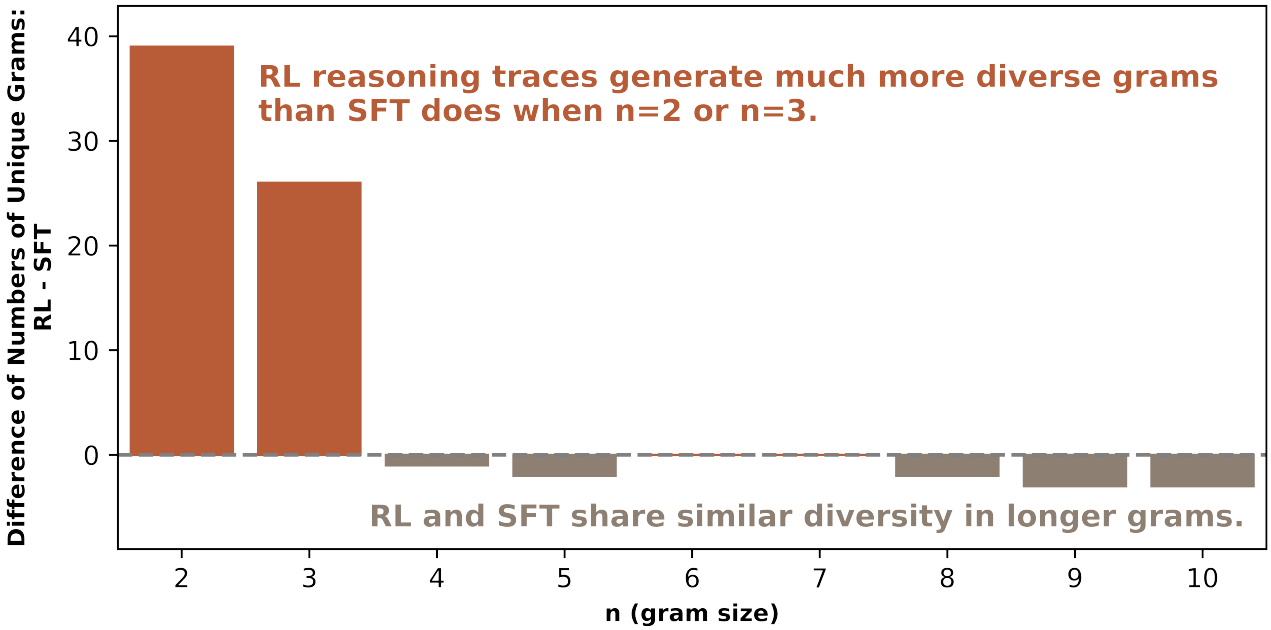

Figure H7. **Difference of numbers of unique grams between the RL and SFT model.** For the n-grams obtained in Appendix D (lines 979-983), we calculate the number of unique grams for both RL and SFT models, then visualize their differences here. The results show that (1) RL reasoning traces generate much more diverse grams than SFT does when $n = 2$ and $n = 3$; (2) RL and SFT reasoning traces share similar diversity in longer grams.